

Jobaer Ahmed Saju

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[ORCID](#) [ResearchGate](#) [Google Scholar](#) [LinkedIn](#) [GitHub](#)

EDUCATION

Ph.D. in Environmental and Water Resources Engineering Aug 2024 – Present

Department of Civil, Structural and Environmental Engineering, University at Buffalo, Buffalo, New York, USA

Advisor: Prof. Dr. Kang Sun

Research: High-Resolution NO_x Emissions Estimation from Satellite Observations for Enhanced Air Quality Monitoring and Health Risk Assessment in the United States.

M.Sc. in Civil Engineering Jan 2018 – Dec 2019

Department of Civil Engineering, Khulna University of Engineering & Technology (KUET), Khulna-9203, Bangladesh

Advisor: Prof. Dr. Q. H. Bari

Thesis: *Monitoring and Source Apportionment of Air Pollutants in Khulna City.*

B.Sc. in Civil Engineering Jan 2013 – May 2017

Department of Civil Engineering, Stamford University Bangladesh, Dhaka-1217, Bangladesh

RESEARCH EXPERIENCE

Graduate Research Assistant Aug 2024 – Present

Department of Civil, Structural and Environmental Engineering, The State University of New York at Buffalo, Buffalo, New York, USA

Research Associate Jun 2022 – Jul 2024

Department of Civil Engineering, Khulna University of Engineering & Technology (KUET), Khulna, Bangladesh

Project: *SCIP Plastics Project, Funded by German Government*

Research Assistant, Jul 2019 – Dec 2019

Consultancy, Research and Testing Services, Department of Civil Engineering, Khulna University of Engineering & Technology, Khulna, Bangladesh

HONORS AND AWARDS

- **NASA FINESST Fellowship (2025–2028):** Awarded \$150,000 by NASA's Earth Science Division to support three years of Ph.D. research titled "*High-Resolution NO_x Emission Estimation From TEMPO Observations: Advancing Air Pollution Monitoring and Public Health Analysis.*"
- **SUNY GREAT Award (2026):** Awarded \$7,500 by *State University of New York (SUNY) system* to support research and professional development.
- **Graduate Research Assistantship (2024–Present):** Department of Civil, Structural and Environmental Engineering, University at Buffalo, The State University of New York, USA.
- **Merit Scholarship:** Academic merit scholarship for three consecutive semesters during the Bachelor's program at Stamford University Bangladesh.

TEACHING EXPERIENCE

Senior Lecturer Apr 2022 – Jun 2022

Department of Civil Engineering, European University of Bangladesh (EUB), Dhaka, Bangladesh

Lecturer Mar 2020 – Mar 2022

Department of Civil Engineering, European University of Bangladesh (EUB), Dhaka, Bangladesh

Courses Taught: Environmental Engineering, Fluid Mechanics, Structural Engineering, Construction and Real

Estate Management, Project Planning and Communication, etc.

Teaching Assistant

Jul 2018 – Jun 2019

Department of Civil Engineering, Khulna University of Engineering & Technology (KUET), Khulna, Bangladesh

INDUSTRY EXPERIENCE

Senior Environmental Engineer

Aug 2020 – Jul 2024

Global Environmental and Textile Technological Services (GETTS), Dhaka, Bangladesh

Job Type: Casual

Responsibilities: Air quality monitoring, assessment, analysis, and modeling; Preparation of Air pollution control system (APCS) design report and presentation.

Design Engineer - APCS

Mar 2021 – Jul 2024

Waste Treatment and Environmental Consultancy (WASTTEC), Dhaka, Bangladesh

Job Type: Casual

Responsibilities: Preparation of EIA, EMP and SIA report; Design and drawing of APCSs; Investigation and drawing of the wastewater treatment plant.

JOURNAL ARTICLES: SUBMITTED

1. Setu, S., Saha, A., **Saju, J. A.**, Bari, Q. H., Mohiuddin, K. A. (2026). *Particulate Matter Pollution and Mortality in Khulna City of Bangladesh: Quantitative Health Risk Assessment Using USEPA and AirQ⁺ Model*. *Heliyon*, Under Review.

JOURNAL ARTICLES: PUBLISHED

11. Ali, M. T., Bari, Q. H., and **Saju, J. A.**. (2025). *Impacts of Natural Climate on California Wildfire Dynamics*. *Scientific Reports*, 2026. <https://doi.org/10.1038/s41598-026-59766-3>
10. Sun, K., **Saju, J. A.**, Nowlan, C. R., Abad, G. G., and Liu, X. (2025). *Hourly Nitrogen Oxides Emissions Estimated From TEMPO and Comparison With Facility-Level Monitoring Data*. *Journal of Geophysical Research: Atmospheres*, Volume 130, Issue 123. <https://doi.org/10.1029/2025JD044565>.
9. Mondal, S., Riaz, A., Nasrin, S. A., Rubel, S. N. R., Setu, S., and **Saju, J. A.** (2024). *Sustainability Evaluation of Municipal Solid Waste to Energy Options for Khulna City, Bangladesh*. *Green Energy and Environmental Technology*, Volume 4, p.1-23. <https://doi.org/10.5772/geet20250073>.
8. **Saju, J. A.**, Bari, Q. H., Lorber, P., Islam, M. R. and Kraft, E. (2024). *Integrated Approach of Waste Analysis and Life Cycle Assessment for the Management of Non-Recyclable Plastics in Recycling Shops*. *Cleaner Environmental Systems*, Volume 15, p.100229. <https://doi.org/10.1016/j.cesys.2024.100229>. **Scopus Indexed (Q1)**, IF: 6.1 (2023).
7. **Saju, J. A.**, Bari, Q. H., Islam, M. R., Alamgir, M., Kraft, E. and Lorber, P. (2024). *Observation of Non-Recyclable Plastic in Recycling Shops: Present Practice and Potential Usage*. *Journal of Material Cycles and Waste Management*, Volume 26, p.800–815. <https://doi.org/10.1007/s10163-023-01880-9>. **Scopus Indexed (Q2)**, IF: 2.7 (2023).
6. Hasan, A. A., Bari, Q. H., Lorber, P., Rafizul, I. M., **Saju, J. A.** and Kraft, E. (2024). *An Association Rule Mining approach to explore the dynamics in plastic recycling business*. *Cleaner Waste Systems*, Volume 9, p.100186. <https://doi.org/10.1016/j.clwas.2024.100186>. **Scopus Indexed (Q2)**, IF: 2.7 (2023).
5. **Saju, J. A.**, Bari, Q. H., Mohiuddin, K. A. and Strezov, V. (2023). *Measurement of Ambient Particulate Matter (PM_{1.0}, PM_{2.5} and PM₁₀) in Khulna City of Bangladesh and Their Implications for Human Health*. *Environmental Systems Research*, 12, 42. <https://doi.org/10.1186/s40068-023-00327-2>. **Scopus Indexed (Editors' Choice article)**.

4. Mazumder, P., Kabir, S. B., **Saju, J. A.**, Islam, M. R. and Emon, A. I. (2023). *Measuring and modeling PM_{2.5} zonal distributions, assembling geospatial and meteorological variables in the Khulna metropolitan area*. *Urban Climate*, Volume 49, p.101518. <https://doi.org/10.1016/j.uclim.2023.101518>. **Scopus Indexed (Q1)**, IF: 6.0 (2023).
3. **Saju, J. A.**, Bari, Q. H. and Mohiuddin, K. A. (2022). *Assessment of gaseous air pollutants motility in Khulna City of Bangladesh using direct sense probes*. *International Journal of Environmental Engineering*, Volume 11(3), p.225–239. <https://doi.org/10.1504/IJEE.2022.124619>. **Scopus Indexed**.
2. **Saju, J. A.**, Bari, Q. H., Nayan, S. B. and Rahman, M. M. (2021). *Qualitative Source Apportionment and Investigation of Long-Range Transport of Air Pollutants*. *Journal of Advanced Civil Engineering Practice and Research*, Volume 12, Page 2–10. **Peer Reviewed**.
1. **Saju, J. A.**, Rubel, S. N. R., Rahman, M. M., Nayan, S. B. and Bagchi, R. (2020). *Effectiveness of manual bar screen in separating solid waste from municipal sewers of Khulna City*. *Australian Journal of Engineering and Innovative Technology*, Volume 2(3), p.31–41. <https://doi.org/10.34104/ajeit.020.031041>. **ISI Indexed**, IF: 1.433 (2023).

EDITED BOOK

- Alamgir, M., Islam, S. M. T., **Saju, J. A.** (2023). *Proceedings of the WasteSafe 2023 – 8th International Conference on Integrated Solid Waste & Faecal Sludge Management*. ISBN: 978-984-35-4034-8. Available at: <https://wastesafe.info/>

POSTER PRESENTATIONS

4. **Saju, J. A.**, Sun, K., Ayazpour, Z. (2026). *A Physics-Based Neural Network Framework for Estimating Spatiotemporal NO_x Reactivity From Satellite Observations*. *Climate Change and Emerging Contaminants: Risks and Responses in Asia and Beyond*, University at Buffalo, April 9–11, 2026.
3. Nazrul, F. B., **Saju, J. A.**, Sun, K. (2026). *Towards Global Modeling of Nitrogen Oxide Chemistry on UB's Super Computers Informed by Satellite Observations*. *Climate Change and Emerging Contaminants: Risks and Responses in Asia and Beyond*, University at Buffalo, April 9–11, 2026.
2. **Saju, J. A.**, Sun, K., Nowlan, C. R., González Abad, G., Liu, X. (2025). *Hourly Nitrogen Oxides Emissions Estimated From TEMPO and Comparison With Facility-Level Monitoring Data*. *TEMPO/GeoXO/ACX Joint Science Team Meeting (Poster P22)*, Center for Astrophysics | Harvard & Smithsonian, Cambridge, MA, Aug 19–21, 2025.
1. Sun, K., **Saju, J. A.**, Ayazpour, Z., Gao, M. (2025). *Learning Temporal and Land Type-Specific Spatial Patterns From Satellite Data-Based Emissions*. *TEMPO/GeoXO/ACX Joint Science Team Meeting*, Center for Astrophysics | Harvard & Smithsonian, Cambridge, MA, Aug 19–21, 2025.

CONFERENCE PAPERS

21. **Saju, J. A.**, Bari, Q. H., Lorber, P., Islam, M. R., Kraft, E., and Hasan, A. A. (2024). *Environmental Benefit of Plastic Recycling in Khulna City of Bangladesh in the Lens of Life Cycle Assessment*. *6th Euro-Mediterranean Conference for Environmental Integration (EMCEI-2024)*, Marrakesh, Morocco, 15–18 May 2024.
20. Paul, P. P., Bari, Q. H., **Saju, J. A.**, and Islam, M. R. (2024). *Life Cycle Assessment of Municipal Solid Waste Management in Khulna City of Bangladesh*. *7th International Conference on Civil Engineering for Sustainable Development (ICCESD 2024)*, Khulna, Bangladesh, pp. 164. ISBN: 978-984-35-5705-6. doi:10.1063/5.0247428
19. Emon, M. E. H., Bari, Q. H., **Saju, J. A.**, Islam, M. R., and Sarker, A. A. (2024). *Classification of Recyclable Plastic Waste by Near-Infrared Spectrometer*. *7th International Conference on Civil Engineering for Sustainable Development (ICCESD 2024)*, Khulna, Bangladesh, pp. 144. ISBN: 978-984-35-5705-6.

18. Lorber, P., Hasan, A. A., Bari, Q. H., Islam, M. R., and **Saju, J. A.** (2024). *Plastic Recycling Business in Khulna: A Study on Mitigating Current Problems*. 7th International Conference on Civil Engineering for Sustainable Development (ICCESD 2024), Khulna, Bangladesh, pp. 178. ISBN: 978-984-35-5705-6.
17. Sarker, A., Bari, Q. H., Islam, M. R., **Saju, J. A.**, Lorber, P., Hasan, A. A., and Farsi, M. M. N. (2024). *Baseline Study for Business Model of Mismanaged Plastic Wastes in Khulna City of Bangladesh*. 7th International Conference on Civil Engineering for Sustainable Development (ICCESD 2024), Khulna, Bangladesh, pp. 187. ISBN: 978-984-35-5705-6.
16. Hasan, A. A., Bari, Q. H., Islam, M. R., Hossain, A., **Saju, J. A.**, Kraft, E., and Lorber, P. (2023). *Socio-Environmental Role of Stakeholders in Recycling Waste Trade in Khulna City*. International Conference on Planning, Architecture and Civil Engineering, RUET, Rajshahi, Bangladesh, pp. 1212–1217.
15. Ahmed, M. K., **Saju, J. A.**, Bari, Q. H., Islam, M. R., Alamgir, M., Kraft, E., and Lorber, P. (2023). *Composition Analysis of Plastic Waste in Recycling Shops of Khulna City in Bangladesh*. WasteSafe 2023 – 8th International Conference on Integrated Solid Waste and Faecal Sludge Management, pp. 133–144. ISBN: 978-984-35-4034-8.
14. Islam, R., **Saju, J. A.**, Noman, A. A., Masudur, M. M., Ahmed, M. A., Podder, J., Rahman, M. A., and Islam, N. (2023). *Monitoring and Assessment of PM_{2.5} and PM₁₀ in Gazipur, Bangladesh*. WasteSafe 2023 – 8th International Conference on Integrated Solid Waste and Faecal Sludge Management, pp. 108. ISBN: 978-984-35-4034-8.
13. Sathi, S., **Saju, J. A.**, Ahmed, M. A., Podder, J., Noman, A. A., Haque, M. E., Azad, M. A. K., and Datta, B. (2023). *Effects of Meteorological Parameters on Primary Air Pollutants: Case Study at Dhaka (CAMS-3, Darus-Salam)*. WasteSafe 2023 – 8th International Conference on Integrated Solid Waste and Faecal Sludge Management, pp. 107. ISBN: 978-984-35-4034-8.
12. Dola, S. H., **Saju, J. A.**, Haque, M. E., Joy Podder, J., Setu, S., Azad, M. A. K., Shila, N., Alam, M. N., and Datta, B. (2023). *Trajectory Analysis of Air Pollutants from February 2020 to January 2022 at a University Campus*. WasteSafe 2023 – 8th International Conference on Integrated Solid Waste and Faecal Sludge Management, pp. 106. ISBN: 978-984-35-4034-8.
11. Imran, M. H., Setu, S., Joy, S., Rahman, M. M., and **Saju, J. A.** (2023). *Evaluation of Rainwater Harvesting System in a Local Pump and Household for Potable Water*. WasteSafe 2023 – 8th International Conference on Integrated Solid Waste and Faecal Sludge Management, pp. 104–105. ISBN: 978-984-35-4034-8.
10. Paul, S., Islam, M. R., Alamgir, M., Sarkar, S. K., Kraft, E., Biastoch, G., Setu, S., and **Saju, J. A.** (2023). *Role of SCIP Plastics Project to Mitigate Plastic Pollution in Khulna Region of Bangladesh*. 19th International Symposium on Waste Management and Sustainable Landfilling – SARDINIA 2023, Cagliari, Italy, 9–13 October 2023.
9. **Saju, J. A.**, Rahman, M. M., Debnath, P. K., and Nayan, S. B. (2020). *Impacts of Air Pollution on Human Health and Environment Due to Brick Kilns Emission: A Review*. 2nd International Conference on Research and Innovation in Civil Engineering – 2020, ISBN: 978-984-3-8047-7, pp. 203–211.
8. Mazumder, P., **Saju, J. A.**, and Bari, Q. H. (2020). *Estimation of Particulate Matter Pollution on Different Major Roadways in Khulna Using Geospatial & Environmental Analysis*. 10th IconSWM–CE 2020, Jadavpur University, Kolkata, India, pp. 75.
7. Islam, R., **Saju, J. A.**, and Rahman, M. M. (2020). *Study on Air Pollution Around a Cement Industry in Khulna City*. 2nd International Conference on Research and Innovation in Civil Engineering – 2020, ISBN: 978-984-3-8047-7, pp. 197–202.

6. Enam, M. S., **Saju, J. A.**, and Bari, Q. H. (2020). *Particulate Matter Concentration Emitted from Heterogeneous Traffic Near Urban Roadways in Khulna City. 5th International Conference on Civil Engineering for Sustainable Development (ICCESD 2020)*, ISBN: 978-984-34-8763-6, pp. 130.
5. Satter, M. A., **Saju, J. A.**, and Islam, M. S. (2020). *Assessment of Particulate Matter and Noise Pollution at Different Road Intersections in Khulna City. 5th International Conference on Civil Engineering for Sustainable Development (ICCESD 2020)*, ISBN: 978-984-34-8763-6, pp. 91.
4. Mazumder, P., **Saju, J. A.**, and Bari, Q. H. (2020). *Commuter Exposure to Air Pollutants in Khulna Railway Station Due to the Influence of Locomotives. 5th International Conference on Civil Engineering for Sustainable Development (ICCESD 2020)*, ISBN: 978-984-34-8763-6, pp. 101.
3. Rubel, S. N. R., Rahman, M. M., and **Saju, J. A.** (2020). *Review Study on Different Concrete Confinement Materials and Mechanisms. 2nd International Conference on Research and Innovation in Civil Engineering – 2020*, ISBN: 978-984-3-8047-7, pp. 105–112.
2. Nayan, S. B., Bari, Q. H., Debnath, P. K., and **Saju, J. A.** (2020). *Performance Study of Pilot-Scale Anaerobic–Aerobic Filter System for Faecal Sludge Treatment. 5th International Conference on Civil Engineering for Sustainable Development (ICCESD 2020)*, ISBN: 978-984-34-8763-6, pp. 132.
1. Debnath, P. K., Bari, Q. H., Nayan, S. B., and **Saju, J. A.** (2020). *Study on the Solids and COD Removal of Shrimp Processing Wastewater Through a Slow Sand Filter in Khulna. 5th International Conference on Civil Engineering for Sustainable Development (ICCESD 2020)*, ISBN: 978-984-34-8763-6, pp. 146.

RESEARCH PROJECTS (BANGLADESH)

- Sustainable Capacity Building to Reduce Irreversible Pollution by Plastics — *SCIP Plastics Project*, Funded by the Federal Ministry for the Environment, Nature Conservation and Nuclear Safety (BMU), Germany. *Role: Lead of Working Group (II): Recycling Shop Assessment.*
- Environmental Quality Evaluation of “Coastal Embankment Improvement Project – Phase I (CEIP-I)”, Assigned to CRTS (Civil), Khulna University of Engineering & Technology. *Role: Research Assistant.*

PERSONAL / ENGINEERING PROJECTS

Air Quality (Design, Report, DoE Presentation)

Air Treatment Plants of Zhongliang New Energy Technology Co. Ltd.; Stellar Industries Co. Ltd.; Shova Power ATP; Safat Battery & Lead Refining Industry; Shamoli Cement Sheet; S.S. Steel Ltd.; M.S. Tania Metal Works; Hi-Tech Steel & Re-Rolling Mills Ltd.; Dong Ao New Electric Co. Ltd.; Bismillah Battery; Jianfuo Battery; Legend Battery.

Other Projects (Team Member)

Effluent Treatment Plants of Bruvana Beverages Ltd. (Narayanganj); Compliance Fashion Ltd. (Gazipur); Zhiyuan Industry Co. Ltd. (Manikganj); Medical College for Women and Hospital (Dhaka); Environmental Impact Assessment (EIA) of Partex Agro Ltd. (Natore); Environmental Management Plan (EMP) of Seba Shushrusha Hospital Ltd. (Gazipur); Initial Environmental Examination (IEE) of Concept Tower (Feni); EIA for the Extension of Jessore Airport (Jessore).

SUPERVISED THESES AT EUB

- Effects of Meteorological Parameters on Primary Air Pollutants: Case Study at Dhaka (CAMS-3, Darus-Salam) for Two Years, 2022.
- Backward Trajectory Analysis of Air Pollutants at European University of Bangladesh (EUB) for One Year (Feb. 2021–Jan. 2022), 2022.
- Study on Air Mass Trajectory Analysis for EUB from February 2020 to January 2021, 2022.

- Planning and Design of the Superstructure of a Two-Storied Mosque, 2022.
- Planning and Design of the Superstructure of a Six-Storied Residential Building, 2022.

WORKSHOPS AND TRAINING

- “Workshop on Geotechnical Engineering for Practicing Civil Engineers” organized by the Department of Civil Engineering, *Khulna University of Engineering & Technology (KUET)* on 10–11 February 2020.
- “Workshop on Post-Disaster Recovery: Theory to Practice” organized by the Institute of Disaster Management, *KUET* on October 22, 2019.
- “Workshop on Research Planning and Design” organized by the Institute of Disaster Management, *KUET* on April 20, 2019.
- “International Seminar on Present Status of Climate Science and Marine Meteorology” organized by the Department of Physics, *KUET* on February 8, 2019.
- “Workshop on Research Integrity and Publication Ethics” organized by the Institutional Quality Assurance Cell (IQAC) of *KUET* on July 14, 2018.

PROFESSIONAL ACTIVITIES

- **Member**, American Geophysical Union (AGU), 2025–Present.
- **Peer Reviewer** for international journals including *Heliyon*, *International Journal of Environmental Science and Technology*, *Sustainable Cities and Society*, *Urban Climate*, and *Waste Management Bulletin*.
- **Associate Member**, The Institution of Engineers, Bangladesh (IEB), since 2017.

EXTRACURRICULAR ACTIVITIES

- **Secretary (2025-2026)**, Graduate Student Association – Department of Civil, Structural and Environmental Engineering (CSEE), University at Buffalo.
- **Executive Member (2025-2026)**, Environmental and Water Resources Institute (EWRI) Chapter, University at Buffalo.
- **Volunteer (2019)**, 6th International Conference on Integrated Solid Waste & Faecal Sludge Management, Khulna, Bangladesh, on February 23-24, 2019.
- **Executive Member (2016–2017)**, Stamford University Sahitto Forum (SUSF), organized inter-university cultural and literary events.

TECHNICAL SKILLS

- **Programming:** Python, MATLAB, R
- **Analysis & Modeling:** openLCA, SPSS, Origin, HYSPLIT
- **Engineering Design:** AutoCAD, ETABS, STAAD.Pro
- **Operating Systems:** Linux, Windows
- **Documentation:** LaTeX, Microsoft Office

LANGUAGES

- Bengali (Native), English: Overall IELTS Score: 7.0 (Listening 7.5, Reading 6.5, Writing 6.0 & Speaking 7.0)

MEDIA REPORTS

- **CSEE Student Awarded NASA Fellowship for Air Quality Monitoring Research**, published by the *Department of Civil, Structural and Environmental Engineering, University at Buffalo*, September 10, 2025. Available online: [UB Engineering News: NASA Fellowship Feature](#)

- **Three SEAS students receive SUNY Great awards**, published by the *Science & Engineering Shared Facilities – University at Buffalo*, June 11, 2026. Available online: [UB Engineering News: SUNY GREAT Award](#)

REFERENCES

- **Prof. Dr. Kang Sun**, Department of Civil, Structural and Environmental Engineering, University at Buffalo, Buffalo, NY, USA, Email: kangsun@buffalo.edu
- **Prof. Dr. Q. H. Bari**, Department of Civil Engineering, Khulna University of Engineering & Technology (KUET), Khulna, Bangladesh, Email: qhbari@ce.kuet.ac.bd
- **Prof. Dr. Kazi A. B. M. Mohiuddin**, Department of Civil Engineering, Khulna University of Engineering & Technology (KUET), Khulna, Bangladesh, Email: kzmohiuddin@ce.kuet.ac.bd